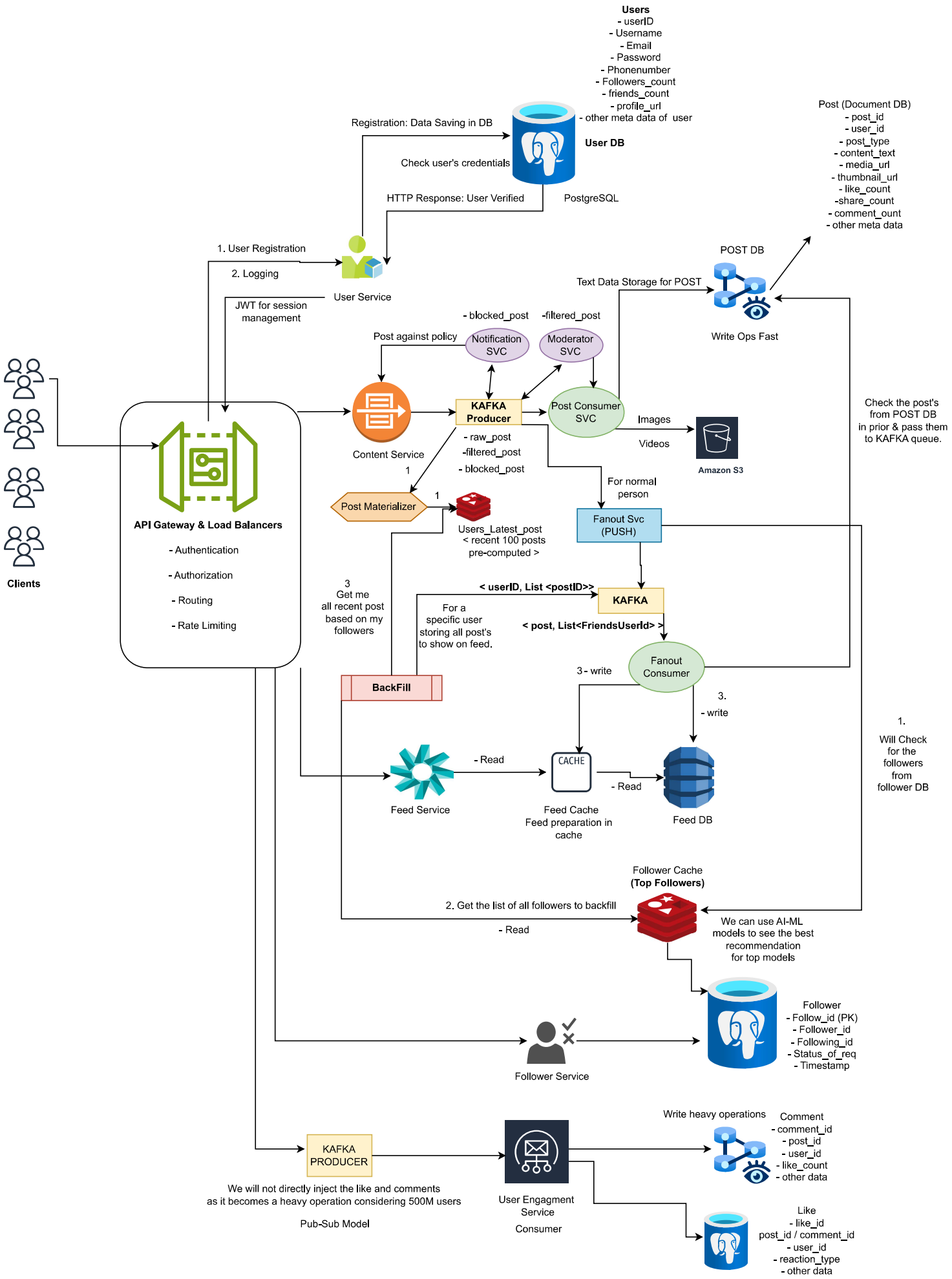
Users
- userID
- Username
- Email
- Password
- Phonenumber
- Followers_count
- friends_count
- profile_url
- other meta data of user
User DB
PostgreSQL
Registration: Data Saving in DB
Check user's credentials
HTTP Response: User Verified
Post (Document DB)
- post_id
- user_id
- post_type
- content_text
- media_url
- thumbnail_url
- like_count
-share_count
- comment_ount
- other meta data
1. User Registration
2. Logging
User Service
JWT for session management
POST DB
Text Data Storage for POST
Write Ops Fast
- blocked_post
-filtered_post
Post against policy
Notification SVC
Moderator SVC
Clients
API Gateway & Load Balancers
- Authentication
- Authorization
- Routing
- Rate Limiting
Content Service
KAFKA Producer
Post Consumer SVC
Images
Videos
Amazon S3
Check the post's from POST DB in prior & pass them to KAFKA queue.
- raw_post
-filtered_post
- blocked_post
1
Post Materializer
1
Users_Latest_post
< recent 100 posts pre-computed >
For normal person
Fanout Svc (PUSH)
3
Get me all recent post based on my followers
< userID, List <postID>>
KAFKA
For a specific user storing all post's to show on feed.
< post, List<FriendsUserId> >
Fanout Consumer
3 - write
3.
- write
BackFill
1.
Will Check for the followers from follower DB
- Read
CACHE
- Read
Feed Service
Feed Cache
Feed preparation in cache
Feed DB
Follower Cache
(Top Followers)
2. Get the list of all followers to backfill
- Read
We can use AI-ML models to see the best recommendation for top models
Follower
- Follow_id (PK)
- Follower_id
- Following_id
- Status_of_req
- Timestamp
Follower Service
Write heavy operations
Comment
- comment_id
- post_id
- user_id
- like_count
- other data
KAFKA PRODUCER
We will not directly inject the like and comments as it becomes a heavy operation considering 500M users
Pub-Sub Model
User Engagment Service
Consumer
Like
- like_id
post_id / comment_id
- user_id
- reaction_type
- other data